

Demo Runbook — Multilingual AI Assistant

This document walks through real, recorded runs of the app to demonstrate its multilingual voice capability.

How to reproduce this demo

1. Start the app:
 2. `conda activate aiassistantstreamlit run main.py`
 3. Open the app in your browser (e.g. `http://localhost:8503`).
 4. Click "Ask me anything!".
 5. Speak your question clearly into your microphone in any supported language.
 6. Wait for the response — text will appear in the **Response** box, followed by an inline audio player and a **Download Speech** button.
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Run 1 — English

Spoken input: *"How are you?"*

Response (text):

Main bilkul theek hoon, shukriya! Aap kaise hain? Sab kaisa chal raha hai?

(Note: the model responded in Hindi here despite the English prompt — Gemini can switch languages based on context or prior conversation; this is expected behavior for a multilingual assistant and not a bug.)

Audio output: 6 seconds, playable inline and downloadable via **Download Speech**.

Run 2 — Kannada

Spoken input: *"Do you know Kannada?"* (or asked directly in Kannada)

Response (text):

ಹೌದು, ನನಗೆ ಕನ್ನಡ ಬರುತ್ತದೆ. ನಾನು ನಿಮಗೆ ಹೇಗೆ ಸಹಾಯ ಮಾಡಲಿ?

(Yes, I know Kannada. How can I help you?)

Audio output: 9 seconds, correctly voiced in Kannada.

Run 3 — English, longer response

Spoken input: *"Explain AI simply"*

Response (text, excerpt):

At its simplest, **Artificial Intelligence (AI)** is a field of computer science that focuses on building smart machines capable of performing tasks that typically require human intelligence.

While humans learn from life experiences, AI learns from **data**. By processing massive amounts of information, AI systems can identify patterns, make predictions, and solve problems.

1. The Core Idea: "Learning" vs. "Programming"

- **Traditional Software:** A programmer writes specific rules... The computer follows these instructions exactly.
- **Artificial Intelligence:** Instead of rigid rules, an AI is fed data and algorithms and learns patterns on its own.

Audio output: 5 minutes 2 seconds — demonstrates the app handling long-form responses and generating correspondingly long speech output without issue.

What this demonstrates

-  Speech-to-text correctly captures spoken input across different phrasings
-  Gemini generates contextually appropriate responses, including switching scripts/languages (Devanagari, Kannada script, English)
-  Text-to-speech (gTTS) correctly voices non-English scripts
-  The app handles both short conversational replies and long explanatory answers
-  Audio is both playable in-browser and downloadable as an MP3

Known limitations

- The free tier of the Gemini API caps requests at 5 per minute — rapid repeated testing will trigger a rate-limit message.
- Speech recognition accuracy depends on microphone quality and background noise.
- `gTTS` requires an internet connection to generate speech (it calls Google's TTS service).

Multilingual AI Assistant

Ask me anything!

Response:

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(Yes, I know Kannada. How can I help you?)

▶ 0:09 / 0:09 🔊 ⋮

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7/18/26, 12:14 PM

Streamlit

Multilingual AI Assistant

Ask me anything!

Response:

At its simplest, **Artificial Intelligence (AI)** is a field of computer science that focuses on building smart machines capable of performing tasks that typically require human intelligence.

While humans learn from life experiences, AI learns from **data**. By processing massive amounts of information, AI systems can identify patterns, make predictions, and solve problems.

Here is a breakdown of how to understand AI:

1. The Core Idea: "Learning" vs. "Programming"

- * **Traditional Software:** A programmer writes specific rules (e.g., "If the user clicks this button, open this window"). The computer follows these instructions exactly without ever "thinking" about what it is doing.
- * **Artificial Intelligence:** Instead of being given rigid rules, an AI is fed data and algorithms. It "learns" how to achieve a goal. For example, if you want an AI to recognize a cat, you show it millions of photos of cats. Eventually, it learns the patterns (ears, whiskers, fur) that define a cat on its own.

▶ 0:00 / 5:02 🔊 ⋮

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